

$$3.2.) \langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

Radim
Trilety

(a) $\mathcal{L} \{1, x, x^2\}$ prostoru V

$$f_1 = 1, f_2 = x, f_3 = x^2$$

první vektor: $\vec{u}_1 = 1$, norma: $\|\vec{u}_1\|^2 = \int_1^5 1 dx = 4$

$$\Rightarrow \|\vec{u}_1\| = 2, \text{normalizace: } e_1 = \frac{1}{2} \cdot 1$$

druhý vektor: Nejprve projekce $\vec{u}_1$: $\langle x, 1 \rangle = \int_1^5 x dx = 12$

$$\langle 1, 1 \rangle = 4$$

Projekce: $\frac{12}{4} = 3 \Rightarrow \vec{u}_2 = x - 3$, norma: $\|\vec{u}_2\|^2 = \int_1^5 (x-3)^2 dx = \frac{16}{3}$

$$\Rightarrow \|\vec{u}_2\| = \frac{4}{\sqrt{3}}, \text{normalizace: } e_2 = \frac{\sqrt{3}}{4} (x-3)$$

třetí vektor: Nejprve $\vec{u}_1, \vec{u}_2$: $\langle x^2, 1 \rangle = \int_1^5 x^2 dx = \frac{124}{3}$

$$\frac{\langle x^2, 1 \rangle}{\langle 1, 1 \rangle} = \frac{31}{3}$$

druhá projekce: $\langle x^2, x-3 \rangle = \int_1^5 x^2(x-3) dx = 32$

$$\langle x-3, x-3 \rangle = \frac{16}{3}$$

~~NA~~ $\frac{\langle x^2, x-3 \rangle}{\langle x-3, x-3 \rangle} = \frac{32}{\frac{16}{3}} = 6$, tedy: $\vec{u}_3 = x^2 - \frac{31}{3} - 6(x-3) =$

$$= x^2 - 6x + \frac{23}{3}, \text{norma: } \|\vec{u}_3\|^2 = \int_1^5 \left(x^2 - 6x + \frac{23}{3}\right)^2 dx = \frac{256}{45}$$

$$\Rightarrow \|\vec{u}_3\| \approx \frac{2,386}{\sqrt{45}}, \text{normalizace: } e_3 \approx \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right)$$

orthonormalni báse: $e_1 = \frac{1}{2}$, $e_2 = \frac{\sqrt{3}}{4}(x-3)$, $e_3 = \frac{3\sqrt{5}}{16}(x^2-6x+\frac{23}{3})$

(b)

$$p(x) = \sum \langle \sin x, e_i \rangle e_i$$

$$\text{Integrály: } \langle \sin x, 1 \rangle = \int_1^5 \sin x \, dx \approx 0,2566$$

$$\langle \sin x, x-3 \rangle = \int_1^5 \sin x (x-3) \, dx \approx -3,4483$$

$$\langle \sin x, x^2-6x+\frac{23}{3} \rangle = \int_1^5 \sin x (x^2-6x+\frac{23}{3}) \, dx \approx -0,2987$$

Podstatení dostaneme:

$$p(x) = ax^2 + bx + c$$

$$a \approx -0,543, b \approx 2,670, c \approx 2,179$$

$$p(x) \approx -0,543x^2 + 2,670x - 2,179$$

$$\| \sin x - p(x) \| = \sqrt{\int_1^5 (\sin x - p(x))^2 \, dx} \approx 0,57$$

(c)

$$\| \sin x - p(x) \| = \sqrt{\int_1^5 (\sin x - p(x))^2 \, dx} \approx \frac{0,36}{2}$$